Franco Lepore - Spring 2026 Intro to Digital Systems (ENGIN-2213-001)

Portfolio Project 1 - Musical Seven Natural Notes Display v1.0:

Functionality:

The main function of this dispositive is to display in a 7 Common Cathode 7-Segment Display the name of the seven natural notes (letter that represents them) in order, turning ON their correspondent switches. This will also display a designated color light from a RGB LED (if it's a correct combination).

The seven natural notes are C, D, E, F, G, A, B.

I assigned one unique combination to represent each note (which will light up specific segments of the display) and the status of the LED, all this represented in the truth table (Active-Low) below:

Decimal	Note	A B C	RGB LED	Red LED	a	b	c	d	e	f	g
0	Off	0 0 0	OFF	OFF	1	1	1	1	1	1	1
1	C	0 0 1	ON (Blue)	OFF	0	1	1	0	0	0	1
2	D	0 1 0	ON (Green)	OFF	1	0	0	0	0	1	0
3	E	0 1 1	ON (Red)	OFF	0	1	1	0	0	0	0
4	F	1 0 0	ON (Cyan)	OFF	0	1	1	1	0	0	0
5	G	1 0 1	ON (Yellow)	OFF	0	0	0	0	1	0	0
6	A	1 1 0	ON (Magenta)	OFF	0	0	0	1	0	0	0
7	B	1 1 1	ON (White)	OFF	0	0	0	0	0	0	0

Common Anode 7-Segment Display **(0 = ON, 1 = OFF)**

RGB LED Color code:

(E) Red = R
(D) Green = G
(C) Blue= B
(F) Cyan = G+B
(G) Yellow = R+G
(A) Magenta = R+B
(B) White = R+G+B

Minimum SOP expressions for each segment: (Incorrect/in development)

- Segment A: C+B
- Segment B = AB’+BC
- Segment C = AB’+BC
- Segment D = AC’+B
- Segment E = A+B’C+BC’
- Segment F = C+B
- Segment g = C+A

Components Used:

- Common Anode 7-Segment Display
- 7408 Quad AND gate (2)
- 7404 Hex Inverter (2)
- 7432 Quad OR gate (2)
- DIP Switch (1)
- 220-ohm resistors (8)
- RGB LED
- Breadboard (1)